

Personal Statement

Is there anything else you would like us to know about you or take into consideration when reviewing your application? Is there something that would give us a more complete picture of who you are as a person? Often this space is used to convey personal anecdotes or a narrative that might provide context elsewhere (i.e., grades, missing information, educational history, how you've overcome adversity, or a personal story/joke) that would be important and relevant to your application. (max 2250 characters)

In all my years of education, I've never had a Latino professor. I often felt that my Mexican identity was contrary to my identity as a scientist. My background and my culture are very important to me, and I want to be able to feel like myself as I pursue a career in research. A PhD will help me change this for the next generation of Latinx computational ecology students, and become the mentor that I wish I had.

My family always impressed upon me the supreme importance of education. My grandparents were refugees, and my father emigrated to the United States from Mexico to pursue higher education. I feel lucky to have grown up in a household that encouraged me to never stop asking questions. My sister and I were taught that we had a responsibility to use our education to give back to the world. I feel uniquely suited to doing this by transforming the field of computational ecology, a field in which Latinos are highly underrepresented.

I have also faced some physical barriers to education. Halfway through my senior year of college, I was hit by a car while crossing the street and suffered multiple injuries. I was incapacitated for a long period of time, and my recovery was difficult. For several months, I was unable to work for more than a couple hours each day, and I relied heavily on the support of my friends, family, and mentors. I considered taking a medical leave or dropping out of school entirely, but ultimately realized how important my education was to me. Finishing my degree while recovering from the accident was incredibly difficult, but I am proud to have persevered and am grateful that I can now continue my education with a PhD. Importantly, I am very grateful to the teachers and friends who helped me through this difficult period, and I want to do everything I can to give back and help others in need.

Throughout my career, I will promote inclusion at every level of teaching and research. Over the past several years, I have worked to reconcile my identity, and am proud to call myself a Latino scientist. I hope to use my education to teach the next generation of computational biologists. Representation is especially important in the field of computer science and its intersection with ecology, wherein adverse effects of climate change disproportionately affect people of color who are underrepresented in STEM. Decisions about conservation and sustainability are often made without the input of the people whose communities they most impact. By pursuing a career in computational ecology, I will be able to contribute to the conversation.

Laboratory and Research Experience/Other Employment

Please include any research or work experience that you feel would be relevant or important information as part of your application. Begin with current or most recent experience: please include organization, start and end dates, position, and one sentence detailing summary of work. (max 4000 characters)

Since starting my PhD this past fall (2023), and as a requirement of my Interdisciplinary Quantitative Biology program, I have completed interdisciplinary rotations in several computational labs across multiple departments. My first rotation was in Dr. Laura Dee's lab, in the department of Ecology and Evolutionary Biology, where I used ordinary differential equations models to simulate population dynamics and predict disturbance responses for rocky shore ecosystems in the Gulf of Maine. I did my second rotation in Dr. Grace Leslie's lab in the ATLAS Creative Technology and Design Institute, where I used publicly-available datasets to study the effect of algorithmically-calculated musical features on listeners' physiology.

Prior to starting my PhD, I worked as a research intern for the Smithsonian MarineGEO Network. During my first two internships at the Smithsonian, first as an REU intern in 2019, and then as a remote intern during 2019-2020, I worked on a project where we used deep convolutional neural networks to study the community dynamics of coral reefs in Belize. That effort led to a first-author paper in *Scientific Reports* (11, 19185, 2021), which was highlighted as an Editor's Choice. During those internships, I also created computational tools as a part of my image-processing pipeline which are still in use at MarineGEO today. In a third internship with MarineGEO, in 2023, I curated data on seagrass ecosystems from over 50 network partners in 20 countries around the world and created a clean dataset for publication. I then analyzed my dataset and contributed to a manuscript on our findings, which will be submitted for publication this spring.

During my undergraduate studies, I sought out as many research opportunities as possible. For my undergraduate thesis, I conducted research with Dr. Ethan Clotfelter on the hydrodynamics and biomechanics of crayfish swimming. I used high-speed video and

developed computer vision programs to study the interaction between crayfish and substrate during their escape response. I continued to analyze the results of this study independently in the year after I graduated, and wrote a second first-authored paper based on my findings, which is now under review in *Integrative Organismal Biology*.

During the summer of 2022, before starting my final semester of undergrad, I did an REU internship in Dr. Carla Restrepo's lab at the University of Puerto Rico Rio Piedras. In the Restrepo lab, I used programs including ArcGIS, ERDAS Imagine, Python, and R, alongside remote sensing data, to study the lithology of the Sierra de Las Minas in Guatemala, with the aim of identifying landsliding risk factors in this difficult-to-access region.

During the fall of 2021, in the middle of the pandemic, I elected to take a semester off from college to pursue a research opportunity with Dr. Mariano Lastra Valdor at the University of Vigo, in Spain. I worked as a research assistant and conducted a study on the relationships between biodiversity, algal wrack, and sediment and water nutrients on Galicia's sandy beaches. I contributed my results and visualizations to a manuscript on our findings, which we plan to submit for publication this winter.

In the summer of 2021, I conducted an internship with Dr. Nipam Patel at the Marine Biological Laboratory, where I developed computational tools, including a computer vision program, to identify patterns in butterfly wing development.

Academic Awards and Honors

Include undergraduate and graduate honors (if applicable). (4000 characters)

During the fall of 2023, on entering grad school, I received a one-year NSF Integrated Data Science Fellowship to support my participation in the Interdisciplinary Quantitative Biology certificate program. I also received a Colorado Diversity Initiative Fellowship from the University of Colorado.

In December of 2022, I graduated from Amherst College *cum laude* with degrees in Biology and Music, as well as a certificate in Coastal and Marine Science.

In the spring of 2020 and summer of 2021, I received two Gregory S. Call Undergraduate Research Awards from Amherst College to support my independent research.

My first paper, entitled “A doubling of stony coral cover on shallow forereefs at Carrie Bow Cay, Belize from 2014 to 2019”, in *Scientific Reports*, 19185 (2021), was highlighted as an Editor’s Choice by the journal.

Extracurricular Activities

Your extracurricular activities can show evidence of leadership and commitment. Please include your involvement in any activity (i.e., technical societies, service organizations, athletics, music, outreach, etc.) that you feel best represents those characteristics. You may also include a description detailing your involvement if you so choose. (4000 characters)

Since starting my PhD, I have participated actively in a number of extracurricular activities. I am a member of QSci, an affinity group for queer students and allies in STEM, and as a Colorado Diversity Initiative (CDI) fellow, I frequently engage with community-building events run by the program.

I also volunteer in the community whenever I can. I recently volunteered as a “professional” judge at a local elementary school science fair, and will be working as a volunteer instructor with Snow School, a program that brings elementary school students into nature to learn about biology, over the next few weeks.

During my undergrad, I was active in many student organizations. I held leadership roles on the Club Soccer and Tennis teams, and was a founding member of the Multicultural Students Union, an affinity group for students with mixed or multicultural identities. I also played guitar and wrote and arranged music for several jazz ensembles, and was a founding member of the Amherst College Afro-Cuban Ensemble. Now, I continue to engage with music as much as possible and play guitar in a jazz fusion band.

Field of Interest and the Role of Computational Science

1. In terms a general audience would understand, describe an important, outstanding scientific or engineering challenge in your field of interest where computational science can play an important role.

Ecosystems provide many benefits that make life possible for humans. These benefits — called ecosystem services — include providing raw materials, regulating natural processes, and supporting recreation, aesthetics, and well-being. Ecosystems are subject to many disturbances, which are becoming increasingly frequent and severe with climate change and human pressures. These disturbances can threaten an ecosystem's integrity, and in turn impact the services it provides. Impacts to ecosystem services can have massive economic, social, and natural consequences — but are hard to predict and manage because ecosystem dynamics are driven by interactions between all the species in the system.

Interactions between species are often difficult or impossible to observe empirically, but they can be simulated computationally. By relying on advanced computational methods, coupled to automated heterogeneous data collection, my dream is to use sophisticated computational simulations to reliably predict and manage the effects of disturbances on ecosystem services before such disturbances even occur.

2. Describe the particular science or engineering problem that you would like to pursue in your research and how you would approach this. What would be the impact on the field and/or on science, engineering and/or society in general if this challenge could be successfully addressed?

I am interested in the long-term effectiveness and resiliency of nature-based solutions — those relying on natural processes — for ecosystem services (e.g. carbon sequestration to mitigate climate change). Understanding how ecosystem services are affected by disturbances is central to the effective implementation of nature-based solutions, yet is in its infancy. To this end, it is necessary to advance knowledge of the

complex relationships between ecosystem properties and resiliency to inform targeted ecosystem management strategies, and enable dynamic responses to climate change.

I would like to develop ordinary differential equation models to study how ecosystem service provisioning is affected by various disturbances, including extinctions and invasions. I will use rocky shores as a model system, since they have relatively few species and are well-studied, and study services like carbon sequestration and water filtration. I aim to understand how the complexity and structure of ecosystems (e.g. size, modularity) determines the resiliency of the services they provide.

Use of Computational Science in your research

1. What is the most complex calculation you have run on a high-performance machine as part of your research experience? Or if you haven't run a high-performance computing system, tell us about the most complex computational problem you have tackled.

During the first semester of my PhD, I implemented an ordinary differential equation model of species' feeding interactions to predict the dynamics of rocky shore ecosystems in the Gulf of Maine. I modeled twelve ecosystems, each containing 50-100 species, over time periods ranging from days to years. Each model was run 100 times with a randomized initial state each time in order to calculate confidence levels. To determine changes in species' biomasses, each species had to be compared to every other species in the system at each time step. The change in biomass was estimated as the sum of the energy gained from photosynthesis and from consuming other species, minus the energy lost to metabolism and to consumers. At first, I relied on an R-based implementation of the model. The simulations ran very slowly and I was unable to get all the answers that I was after, so I re-implemented the model on my own, using C. With my implementation, I am now able to run my simulations over one order of magnitude faster on the University of Colorado's supercomputing cluster. I am now working on a CUDA based version of my model, which will allow me to gain additional computational advantages by running on GPUs.

2. Imagine if you were given access to resources 100 times more powerful than what you have access to. What would that enable you to do, and what do you perceive the mathematical and computer science challenges to be?

With such a system I would study much larger and more complex ecosystems. My research is limited to relatively small systems, with a limited number of interactions due to computational limitations. However, I am interested in studying systems that include seagrass meadows and coral reefs, which can have many more species and are more often actively managed than the systems I can study currently. In the same vein, I would like to extend my study to a much wider geographic area to see how my results scale with different climate types and species compositions.

Some of the most exciting work in computational ecosystem dynamics is currently being pursued at the DOE's national laboratories, including Berkeley, PNNL and Argonne. I am very interested in collaborating with those research groups and utilizing their computational resources. I understand that the CSGF Fellowship offers the opportunity to pursue a research internship at a DOE national laboratory, I would be extraordinarily fortunate to be granted such a chance.

Program of Study

Describe how the courses listed in your planned program of study would help prepare you to address the challenges you have described in questions 1 and 2. Discuss your rationale for choosing these courses.

For science applications, I chose causal inference and bioinformatics. In causal inference, I will learn about cutting-edge statistical methodology and its applications to my field. Causal inference is often used in the social sciences, but has rarely been applied to ecological problems. These techniques will prepare me to conduct more ambitious studies, and to work with bigger and messier data. In bioinformatics, I developed an understanding of the theoretical

basis for sequencing technologies and built practical skills that I plan to apply to my research, like metagenome sequencing.

For math and statistics, I chose spatial statistics and statistical learning. Spatial statistics will teach me techniques for working with spatially explicit data, like remote sensing data and photogrammetry data. These technologies provide much larger and richer datasets than have existed before, and bringing them into my research is essential to studying the global problems, like climate change, that I am interested in. In statistical learning, I will learn about the basis for machine learning models. I use machine learning often in my research, so this course will help me better apply existing models and enable me to develop my own new models.

For high-performance computing, I chose high-performance scientific computing. From this course, I will learn about the intricacies of high-performance computing and how to write programs to run efficiently on my University's computing clusters. This will let me handle larger and more complex data, which is one of my primary goals.

For computer science, I chose software engineering for scientists and network analysis. In software engineering, I learned best practices for scientific software development. This course taught me how to collaboratively develop a piece of scientific software and publish it so that other scientists will use it. In network analysis, I hope to develop an intuition for working with network data. Ecosystems are modeled as networks, and many techniques to evaluate them come from network science. This course will teach me new tools for working with networks computationally and give me a base for doing many kinds of computational work, including machine learning.